

Independent Derivation of the General Closed Form of Power Sums

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Abstract

This paper presents an independent derivation of the general closed form for sums of the form, $1^p + 2^p + \dots + n^p$ for some $p \in [0, \infty) \cap \mathbb{Z}$. The derivation employs methods for solving recurrence relations, discovered through independent research, together with the family of functions introduced in my previous work, *An Investigation of a Family of Polynomial-Exponential Product Functions*. The derivation presented here was developed independently without referencing existing mathematical literature.

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1 Introduction

1.1 Foreword

1.1.1 Independency of this derivation

Many methods I use in this paper are those which I have discovered independently. Consequently, the notation used may differ from conventional notation used for these methods. To ensure clarity, I will explicitly define the notation used throughout the paper and provide justification for methods that I have developed independently.

1.1.2 Derivation sub-sections

The derivation is divided into two subsections. The first is included solely for pedagogical purposes; readers who are interested only in the general derivation may proceed directly to the second subsection.

1.2 Notation

1.2.1 Power Sums

We define a power sum as a sum having the following form:

$$\sum_{r=1}^n r^p = 1^p + 2^p + 3^p + \cdots + n^p$$

where

- $n \in \mathbb{N}$ represents the number of terms in the sum,
- $p \in [0, \infty) \cap \mathbb{Z}$ represents the power each term is raised to.

We will denote these sums using $\sigma_p(n)$, so that

$$\sigma_p(n) = \sum_{r=1}^n r^p.$$

1.2.2 n -th derivatives

We will denote the n -th derivative of a function, $f(x)$, as $f^n(x)$ for any $n \in [0, \infty) \cap \mathbb{Z}$. This is not to be confused with function composition, which is not a relevant concept throughout this paper.

1.2.3 Indexes

We will generally use i , j , k as primary, secondary, and tertiary indexes respectively. For example, a nested sum may have the following form:

$$\sum_{i=0}^n \sum_{j=0}^i \sum_{k=0}^j a_{i,j,k}$$

It is imperative that i is not confused with the imaginary unit. To avoid unnecessary confusion, these 3 letters will only be used to represent indexes throughout the paper. However, other letters may be used to represent indexes, such as the r used when defining power sums, to follow general convention for certain objects, improving clarity.

1.2.4 $1/n!$

For convenience, we adopt the notation that $\frac{1}{n!} = 0$ for $n \in (-\infty, 0) \cap \mathbb{Z}$. This convention is used to solely simplify the notation of summations present throughout this paper.

1.2.5 ρ -functions

We define the ρ -functions as the family of single-variable functions whose n -th derivative, evaluated at 0 is equal to $(m + n)^p$ for some non-negative integers m and p . Thus, a ρ -function satisfies

$$f^n(0) = (m + n)^p$$

where

- $m, n, p \in [0, \infty) \cap \mathbb{Z}$.

In my previous work, *An Investigation of a Family of Polynomial-Exponential Product Functions*, I showed (whilst using slightly different notation) that these functions have the following general structure:

$$\rho_p^m : \mathbb{R} \rightarrow \mathbb{R}, \quad \rho_p^m(x) = e^x \sum_{i=0}^p \left(\frac{x^i}{i!} \sum_{j=0}^i (-1)^j \binom{i}{j} (m + i - j)^p \right)$$

So that, $\rho_p^{m+n}(0) = (m + n)^p$. It may seem odd that $m + n$ is used to describe how many times ρ_p has been differentiated, when this could be described with a single parameter; however, this makes the explicit distinction between the initial m differentiations used to obtain ρ_p^m and the subsequent n differentiations when evaluating the defining property of the function.

Throughout this paper, we will always initially set $m = 0$. Consequently, the ρ -functions used in this paper will hereafter be initially defined as follows:

$$\rho_p : \mathbb{R} \rightarrow \mathbb{R}, \quad \rho_p(x) = e^x \sum_{i=0}^p \left(\frac{x^i}{i!} \sum_{j=0}^i (-1)^j \binom{i}{j} (i - j)^p \right)$$

So that, $\rho_p^n(0) = n^p$. However, the ρ_p^m equation may be used to handle derivatives of ρ_p .

2 Derivation

2.1 Specific case example

Consider the following power sum:

$$\sigma_1(n) = 1 + 2 + 3 + \cdots + n.$$

This can be described by the following recurrence relation:

$$\sigma_1(n) = \sigma_1(n-1) + n,$$

with the initial condition that $\sigma_1(0) = 0$. Now, suppose we have a function, $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying the differential equation

$$f'(x) = f(x) + \rho_1'(x), \quad x \in \mathbb{R}.$$

Differentiating this equation $n-1$ times gives

$$f^n(x) = f^{n-1}(x) + \rho_1^n(x).$$

Therefore,

$$f^n(0) = f^{n-1}(0) + \rho_1^n(0).$$

Since $\rho_1^n(0) = n$, we obtain

$$f^n(0) = f^{n-1}(0) + n.$$

If we impose the initial condition that $f(0) = 0$, then $f^n(0)$ satisfies the same recurrence relation and initial condition as $\sigma_1(n)$. Hence,

$$f^n(0) = \sigma_1(n),$$

for all $n \in \mathbb{N}$. Let's reconsider our differential equation,

$$f'(x) = f(x) + \rho_1'(x).$$

Now, assume the complementary function has the form $f_{CF}(x) = Ae^{mx}$ for some $A, m \in \mathbb{R}$. Therefore,

$$mAe^{mx} - Ae^{mx} = 0,$$

giving us the auxiliary equation,

$$m - 1 = 0 \implies m = 1,$$

so

$$f_{CF}(x) = Ae^x.$$

Now, we can determine that

$$\rho_1(x) = xe^x \implies \rho_1'(x) = e^x + xe^x.$$

Now, if we consider the trial particular integral, $f_{PI}(x) = (\lambda_1 + \lambda_2 x)e^x$, we see that the $\lambda_1 e^x$ term belongs to the complementary function. Consequently, we multiply this trial PI by x , giving us a new trial PI:

$$f_{PI}(x) = (\lambda_1 x + \lambda_2 x^2) e^x.$$

We can now obtain

$$f'_{PI}(x) = (\lambda_1 + 2\lambda_2 x)e^x + (\lambda_1 x + \lambda_2 x^2) e^x = (\lambda_1 + 2\lambda_2 x)e^x + f_{PI}(x).$$

Substituting f_{PI} and f'_{PI} into the differential equation gives us

$$(\lambda_1 + 2\lambda_2 x)e^x + f_{PI}(x) - f_{PI}(x) = \rho'_1(x),$$

therefore,

$$(\lambda_1 + 2\lambda_2 x)e^x = (1 + x)e^x.$$

Comparing coefficients gives us $\lambda_1 = 1, \lambda_2 = \frac{1}{2}$. Thus, we obtain

$$f_{PI}(x) = \left(x + \frac{1}{2}x^2\right) e^x.$$

$f(x) = f_{CF}(x) + f_{PI}(x)$, so we can determine that

$$f(x) = \left(A + x + \frac{1}{2}x^2\right) e^x$$

is the general solution to our differential equation. Now, if $f^n(0) = \sigma_1(n)$ and $\sigma_1(0) = 0$, then it is necessary that $f(0) = 0$. Therefore, we obtain that $A = 0$, giving us the particular solution to our differential equation:

$$f(x) = \left(x + \frac{1}{2}x^2\right) e^x.$$

We know that e^x can be expanded into the following Maclaurin series:

$$e^x = \frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Therefore,

$$f(x) = \left(x + \frac{1}{2}x^2\right) \left(\frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right).$$

Expanding gives us

$$f(x) = \left(\frac{1}{0!}\right)x + \left(\frac{1}{1!} + \frac{1}{2 \cdot 0!}\right)x^2 + \left(\frac{1}{2!} + \frac{1}{2 \cdot 1!}\right)x^3 + \cdots.$$

Consequently, we can express $f(x)$ as an infinite summation:

$$f(x) = \sum_{i=0}^{\infty} \left(\frac{1}{i!} + \frac{1}{2 \cdot (i-1)!}\right) x^{i+1}.$$

Therefore,

$$f'(x) = \sum_{i=0}^{\infty} (i+1) \left(\frac{1}{i!} + \frac{1}{2 \cdot (i-1)!}\right) x^i,$$

and

$$f''(x) = \sum_{i=1}^{\infty} (i+1)(i) \left(\frac{1}{i!} + \frac{1}{2 \cdot (i-1)!}\right) x^{i-1}.$$

Observing the pattern, we can determine

$$f^n(x) = \sum_{i=n-1}^{\infty} \frac{(i+1)!}{(i+1-n)!} \left(\frac{1}{i!} + \frac{1}{2 \cdot (i-1)!} \right) x^{i+1-n}$$

for $n \in \mathbb{N}$. Thus, we can obtain

$$f^n(0) = \sum_{i=n-1}^{\infty} \frac{(i+1)!}{(i+1-n)!} \left(\frac{1}{i!} + \frac{1}{2 \cdot (i-1)!} \right) (0)^{i+1-n} = \frac{n!}{0!} \left(\frac{1}{(n-1)!} + \frac{1}{2 \cdot (n-2)!} \right),$$

as all other terms of the summation are multiplied by 0. Therefore,

$$f^n(0) = n! \left(\frac{2n + n(n-1)}{2 \cdot (n!)} \right) = \frac{n^2 + n}{2} = \frac{n(n+1)}{2}, \quad \forall n \in \mathbb{N}.$$

As $f^n(0) = \sigma_1(n)$, this gives us the following closed form for the power sum, $\sigma_1(n)$:

$$\sigma_1(n) = \frac{n(n+1)}{2}.$$

It is very well known that, as expected, this is the closed form of $\sigma_1(n)$, so we know that this is correct. This seems a very convoluted method to derive such a simple result; however, this method becomes more useful for deriving a general closed form for $\sigma_p(n)$, $\forall p \in [0, \infty) \cap \mathbb{Z}$.

2.2 General case

Consider any power sum:

$$\sigma_p(n) = 1^p + 2^p + 3^p + \cdots + n^p, \quad p \in [0, \infty) \cap \mathbb{Z}.$$

This can be described by the following recurrence relation:

$$\sigma_p(n) = \sigma_p(n-1) + n^p,$$

with the initial condition that $\sigma_p(0) = 0$. Now, suppose we have a function, $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying the differential equation

$$f'(x) = f(x) + \rho_p'(x), \quad x \in \mathbb{R}.$$

Differentiating this equation $n-1$ times gives

$$f^n(x) = f^{n-1}(x) + \rho_p^n(x).$$

Therefore,

$$f^n(0) = f^{n-1}(0) + \rho_p^n(0).$$

Since $\rho_p^n(0) = n^p$, we obtain

$$f^n(0) = f^{n-1}(0) + n^p.$$

If we impose the initial condition that $f(0) = 0$, then $f^n(0)$ satisfies the same recurrence relation and initial condition as $\sigma_p(n)$. Hence,

$$f^n(0) = \sigma_p(n),$$

for all $n \in \mathbb{N}$. Let's reconsider our differential equation,

$$f'(x) = f(x) + \rho_p'(x).$$

Now, we assume the complementary function has the form $f_{CF}(x) = Ae^{mx}$ for some $A, m \in \mathbb{R}$. Therefore,

$$mAe^{mx} - Ae^{mx} = 0,$$

giving us the auxiliary equation,

$$m - 1 = 0 \implies m = 1,$$

so

$$f_{CF}(x) = Ae^x.$$

We know that

$$\rho_p^m(x) = e^x \sum_{i=0}^p \left(\frac{x^i}{i!} \sum_{j=0}^i (-1)^j \binom{i}{j} (m + i - j)^p \right),$$

therefore,

$$\rho_p'(x) = e^x \sum_{i=0}^p \left(\frac{x^i}{i!} \sum_{j=0}^i (-1)^j \binom{i}{j} (1 + i - j)^p \right).$$

Now, let's consider the following trial particular integral:

$$f_{PI}(x) = (\lambda_1 x + \lambda_2 x^2 + \cdots + \lambda_{p+1} x^{p+1}) e^x.$$

Consequently, we obtain that

$$f'_{PI}(x) = (\lambda_1 + 2\lambda_2x^2 + \cdots + (p+1)\lambda_{p+1}x^p) e^x + f_{PI}(x).$$

Substituting f_{PI} , f'_{PI} , and ρ'_p into our differential equation gives us

$$(\lambda_1 + 2\lambda_2x + \cdots + (p+1)\lambda_{p+1}x^p) e^x = e^x \sum_{i=0}^p \left(\frac{x^i}{i!} \sum_{j=0}^i (-1)^j \binom{i}{j} (1+i-j)^p \right).$$

We can clearly see that both sides are polynomials of degree p , and that we can simply match up the coefficients of each x^{i-1} term on the left to the corresponding coefficients on the right. Thus, we obtain

$$i\lambda_i = \frac{1}{(i-1)!} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p,$$

therefore,

$$\lambda_i = \frac{1}{i!} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p.$$

To shorten writing, lets define the following family of functions:

$$\lambda_i(p) = \frac{1}{i!} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p.$$

Returning the value for λ_i , for any p . Therefore, we can determine that

$$f_{PI}(x) = (\lambda_1(p)x + \lambda_2(p)x^2 + \cdots + \lambda_{p+1}(p)x^{p+1}) e^x = e^x \sum_{i=1}^{p+1} \lambda_i(p) \cdot x^i.$$

$f(x) = f_{CF}(x) + f_{PI}(x)$, so we can determine that

$$f(x) = Ae^x + e^x \sum_{i=1}^{p+1} \lambda_i(p) \cdot x^i$$

is the general solution to our differential equation. Now, if $f^n(0) = \sigma_p(n)$ and $\sigma_p(0) = 0$, then it is necessary that $f(0) = 0$. Therefore, we obtain that $A = 0$, giving us the particular solution to our differential equation:

$$f(x) = e^x \sum_{i=1}^{p+1} \lambda_i(p) \cdot x^i.$$

e know that e^x can be expanded into the following Maclaurin series:

$$e^x = \frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Therefore,

$$f(x) = \left(\frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \right) (\lambda_1(p)x + \lambda_2(p)x^2 + \cdots + \lambda_{p+1}(p)x^{p+1}).$$

Expanding gives us

$$f(x) = \left(\frac{\lambda_1(p)}{0!}\right)x + \left(\frac{\lambda_1(p)}{1!} + \frac{\lambda_2(p)}{0!}\right)x^2 + \cdots + \left(\frac{\lambda_1(p)}{p!} + \frac{\lambda_2(p)}{(p-1)!} + \cdots + \frac{\lambda_{p+1}(p)}{0!}\right)x^{p+1} + \cdots \\ \cdots + \left(\frac{\lambda_1(p)}{(p+1)!} + \frac{\lambda_2(p)}{p!} + \cdots + \frac{\lambda_{p+1}(p)}{1!}\right)x^{p+2} + \cdots.$$

Consequently, we can express $f(x)$ as an infinite summation:

$$f(x) = \sum_{i=1}^{\infty} x^i \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(i-j)!}.$$

Therefore,

$$f'(x) = \sum_{i=1}^{\infty} (i)x^{i-1} \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(i-j)!}, \\ f''(x) = \sum_{i=2}^{\infty} (i)(i-1)x^{i-2} \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(i-j)!}.$$

Observing the pattern, we can determine

$$f^n(x) = \sum_{i=n}^{\infty} \frac{i!}{(i-n)!} x^{i-n} \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(i-j)!}$$

for $n \in \mathbb{N}$. Thus, we can obtain

$$f^n(0) = \sum_{i=n}^{\infty} \frac{i!}{(i-n)!} (0)^{i-n} \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(i-j)!} = \frac{n!}{0!} \sum_{j=1}^{p+1} \frac{\lambda_j(p)}{(n-j)!},$$

as all other terms of the infinite summation are multiplied by 0. We now treat j as a primary index, so replace it with i . Recall that

$$\lambda_i(p) = \frac{1}{i!} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p.$$

Therefore,

$$f^n(0) = \sum_{i=1}^{p+1} \frac{n!}{i!(n-i)!} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p.$$

But, we know that

$$\frac{n!}{i!(n-i)!} = \binom{n}{i},$$

therefore,

$$f^n(0) = \sum_{i=1}^{p+1} \binom{n}{i} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p, \quad \forall n \in \mathbb{N}.$$

As $f^n(0) = \sigma_p(n)$, this gives us the following general closed form for any power sum, $\sigma_p(n)$:

$$\sigma_p(n) = \sum_{i=1}^{p+1} \binom{n}{i} \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} (i-j)^p,$$

where

- $p \in [0, \infty) \cap \mathbb{Z}$,
- $n \in \mathbb{N}$.